Chiharu Mamiya

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Professional Summary

Power systems engineer (GE Vernova, Siemens) specializing in electricity markets, battery storage economics, and energy analytics. Combines hands-on grid simulation experience (PSS/E, PowerFactory) with Python automation and data analysis, now pursuing the EIT InnoEnergy SENSE double-degree MSc at KTH Royal Institute of Technology. Trilingual (English, Japanese, German) with work experience across the US, Germany, and Sweden.

Education

Bachelor of Science **Union College** *Schenectady, NY*

- Major in Electrical Engineering, Minor in Biomedical Engineering, GPA: 3.8
- Honors: Tau Beta Pi Engineering Honor Society, JASSO Scholar, Grew Bancroft Scholar UAS7 Scholar, NYSTAR Scholar
- Coursework: Digital Design, Internet of Things, Embedded Microcontrollers, Control Systems, Semiconductor System

Master of Science **KTH Royal Institute of Technology** *Stockholm, Sweden*

- Major in Smart Electrical Networks and Systems (EIT InnoEnergy SENSE); Year 2 at KU Leuven
- Honors: Diversity, Equity and Inclusion Scholarship
- Coursework: Electricity Pricing and Emissions, Power System Analysis, Power Electronics, The Sustainable Electric Power Engineer

Engineering Experience

Grid Solutions Graduate Engineer **GE Vernova** *Saratoga Springs, NY* **10/2025 – 05/2026**

- Performed short-circuit, steady-state, and transient analyses of MV transmission and distribution networks in PowerFactory for utility and industrial clients, supporting grid modernization studies.
- Developed Python automation for PowerFactory workflows, accelerating short-circuit studies and customer transmission data analysis.
- Built the FACTSFLEX product demonstration interface (HTML, CSS, Django), used by the presales team in customer-facing solution presentations.

Technical Consultant Intern **Siemens** *Fremont, CA* **06/2025 – 08/2025**

- Enhanced ACCC Python tools to streamline contingency and power flow analysis in PSS/E, increasing efficiency by up to 70%.
- Optimized TSAT plotting tool runtime by 94% through code refactoring and parallel processing using optimized libraries, significantly enhancing simulation performance.
- Developed test suites and process flowcharts in Microsoft Visio to support debugging, regression testing, and onboarding, improving overall code reliability.

Hardware Research Intern **FH Münster** *Münster, Germany* **09/2024 – 11/2024**

- Designed and tested schematics on Red Pitaya boards, improving quantum sensing FPGA timing performance by 10%.
- Debugged hardware using VHDL/Verilog in a Linux environment, reducing signal errors, and increasing system uptime.

Software Developer Intern **Siemens** *Schenectady, NY* **06/2024 – 08/2024**

- Automated XML data extraction via Python scripts, reducing manual workload by 25% and pushing updates to GitHub.
- Migrated legacy C++ modules to Fortran and MFC, enhancing cross-platform integration across x86-based infrastructure.
- Communicated progress in daily stand-ups, contributing to 2-week early project delivery.

IT Intern **Grew Bancroft** *Remote* **04/2024 – 08/2024**

- Enhanced React/Node.js application frontend, reducing page load times by 25% and improving UI responsiveness.
- Automated user testing with Jest, streamlining QA workflows and reducing manual testing by 20%.

IoT Student Researcher **Union College** *Schenectady, NY* **10/2023 – 08/2024**

- Processed and labeled 1,200+ motion datasets under IRB protocols, supporting ethical sports biomechanics research.
- Trained SVM/KNN models in Python, achieving 80% classification accuracy for expert vs. novice athlete skills.
- Co-authored paper accepted by IEEE AIIoT; collaborated with 4 professors and 1 peer to iterate ML and experimental design.

Systems Engineering Intern **Medtronic** *Fridley, MN* **06/2023 – 08/2023**

- Developed Python system testing automation scripts with Appium and ADB, boosting embedded system testing efficiency by 30%.
- Conducted system tests on a closed-loop brain stimulation device using oscilloscopes and function generators.
- Presented findings in weekly meetings, helping resolve hardware-software issues faster.

Projects

Capstone Project: Smart Microgrid Node, *Senior Thesis Capstone Project* **04/2025 – 11/2025**

- Developing a solar-powered, scalable, and cost-effective microgrid node integrating IoT sensors and STM32 microcontrollers.
- Designing hardware and firmware for real-time energy monitoring, load balancing, and peer-to-peer communication.
- Implementing communication protocols for decentralized energy data exchange and optimization using Python.

IoT and Gesture Recognition System with Neural Network Deployment on ESP32, *Internet of Things* **05/2024 – 06/2024**

- Developed an agriculture IoT platform with Bluetooth, WiFi, and AWS Core, increasing sensor range by 50 meters.
- Achieved 96% gesture recognition accuracy using TensorFlow and MPU6050, validated across 4 gestures.
- Deployed real-time control on ESP32 boards, enabling live RC car navigation via remote control gestures.

Skills

- **Power systems:** PSS/E, PowerFactory, TSAT, pandapower, contingency & power-flow analysis, short-circuit studies
- **Programming & data:** Python (pandas, NumPy, scikit-learn), Git/GitHub, MATLAB/Simulink, SQL basics, C/C++
- **Markets & finance (developing):** Nord Pool / ENTSO-E data analysis, LCOE & project cash-flow modeling, DSCR/IRR analysis
- **Languages:** English (native) · Japanese (native) · German (B1 in progress) · Swedish (beginner) · Mandarin (elementary)

Additional

Event Manager **InnoEnergy Community+** *Stockholm, Sweden* **08/2026 – Present**

- Designing and coordinating student events, from social gatherings to planned company and industry site visits.

Founder **Beyond Fluent** *Tokyo, Japan* **09/2025 – Present**

- Grew a 5,200+ follower educational community with end-to-end content, analytics, and client coaching